Physical Chemistry Cume

1) From the laws of Thermodynamics, show why a spontaneous reaction has a negative ΔG (**20 points**).

2) Why is the molar heat capacity of $CO_{2(g)}$ higher than $Ar_{(g)}$? (**15 points**)

3) What is the ΔG of $CO_{2(g)}$ as it expands (i) reversibly and (ii) irreversibly from 1L to 2L at 300K, if the initial pressure is 4atm. Assume ideal behavior **(15 points)**

4) A protein undergoes a phase transition from native to denatured state at 50°C with an enthalpy change, ΔH, of 50 $kJmol^{-1}$. The molar heat capacity of the native state is 13 $JK^{-1}mol^{-1}$ lower than the denatured state. What is the ΔG for the transition from the native to denatured state at (i) 25°C and (ii) 50°C? State any assumptions you make. **(50 points)**